

■ 2.4.2 Any Expression Can Be a Head

In most cases, you want the head f of an expression like $f[x]$ to be a single symbol. There are, however, some situations where it is convenient to use more complicated expressions as heads.

Mathematica allows you to use any expression as a head.

Here $a+b$ is the head of the expression.

```
In[1]:= (a + b)[x]
Out[1]= (a + b)[x]
```

This has $f[p]$ as a head.

```
In[2]:= f[p][x]
Out[2]= f[p][x]
```

You can use this kind of expression to give “indexed functions” as heads.

```
In[3]:= f[3][x, y]
Out[3]= f[3][x, y]
```

When you give complicated expressions as heads, you must make sure to put in the appropriate parentheses.

```
In[4]:= ((1 + a)(1 + b))[x]
Out[4]= ((1 + a)(1 + b))[x]
```

One situation in which we have already encountered the use of complicated expressions as heads is in working with pure functions (see Section 2.1.12). If you give `Function[args, body]` as the head of an expression, then the function is applied to any arguments you supply.

With the head `Function[x, x2]`, the value of the expression is the square of the argument.

```
In[5]:= Function[x, x^2][a + b]
Out[5]= (a + b)^2
```

Another important use of more complicated expressions as heads is in implementing *functionals* and *functional operators* in mathematics.

As one example, consider the operation of differentiation. As will be discussed in Section 3.5.4, an expression like f' represents a *derivative function*, obtained from f by applying a functional operator to it. In *Mathematica*, f' is represented as `Derivative[1][f]`: the “functional operator” `Derivative[1]` is applied to f to give another function, represented as f' .

This expression has a head which represents the application of the “functional operator” `Derivative[1]` to the “function” f .

```
In[6]:= f'[x] // FullForm
Out[6]//FullForm= Derivative[1][f][x]
```

You can replace the head f' by another head, say fp . This effectively takes fp to be a “derivative function” obtained from f .

```
In[7]:= % /. f' -> fp
Out[7]= fp[x]
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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